Onkar Jadhav

Machine Learning Engineer
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B.Tech Computer Science, PCCOE (9.08 CGPA)

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Summary

Final-year CS student (9.08 CGPA) building deep learning systems and Python ML pipelines. Built Lore Engine - a full-stack RAG platform combining BM25 lexical search and pgvector dense embeddings via reciprocal rank fusion over 6,000+ pages, with a FastAPI backend, deploy-gated eval harness, and streaming chat interface. Shipped Stride (5–10 active users) with five domain agents, NLP extraction from multi-modal input, and multi-provider LLM routing via OpenRouter. Built PromptForge, an agentic LLM orchestration pipeline with cross-model benchmarking across OpenAI, Anthropic, Gemini, and open-source models. Strong in Python, RAG pipelines, NLP, vector databases, and REST API development.

Education

•Bachelor of Technology in Computer Science *Nov 2022 – Jun 2026*
Pimpri Chinchwad College of Engineering, Savitribai Phule Pune University CGPA: 9.08 (up to 7th semester)

– Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object-Oriented Programming

Experience

•Stride - AI-First Adaptive Wellness Platform *April 2026 – Present*
Founder & Lead Engineer Remote

– Built and shipped Stride to production as sole engineer; running live with **5–10 active trial users** driving iterative product feedback.
– Designed five specialised domain agents with LLM routing via OpenRouter (Anthropic, OpenAI, Gemini) and an NLP extraction layer converting multi-modal natural-language input into structured data without rigid templates.
– Built REST API endpoints integrating generative AI with multi-modal input capture (text, voice, image, barcode, nutrition labels), demonstrating full-stack LLM application development with real-world data integration.

Engineering Projects

•StockXpert - *Full-Stack RAG Platform with Hybrid BM25 + pgvector Retrieval* *2026 – Present*
Next.js, FastAPI, PostgreSQL, pgvector, BM25, Celery, Redis, Cloudflare R2, OpenAI, Anthropic Claude

– Architected a full-stack RAG application ingesting **6,000+ MediaWiki pages** into a hybrid retrieval pipeline (BM25 + pgvector via reciprocal rank fusion) - LangChain-equivalent hybrid retrieval built directly on provider APIs for full control over the retrieval stack.
– Built a FastAPI backend serving streaming LLM responses with inline citations, integrated with a Next.js frontend for real-time chat, and a hierarchically generated reference book with aggressive per-series caching.
– Established a **50-question per-series ground-truth eval harness** measuring retrieval MRR and end-to-end answer accuracy, gating every deployment; scaled to **∼5M vectors per series** in pgvector.
– Built an async ingestion pipeline (Celery, Redis, Cloudflare R2) that fetches from the MediaWiki API, converts wikitext to markdown, and indexes into BM25 and pgvector stores automatically.

•PromptForge - *Agentic LLM Orchestration Pipeline with Cross-Model Benchmarking* *Jan 2026 – Feb 2026*
Node.js, Express.js, BullMQ, Convex, OpenRouter, React.js

– Architected a creator–critic–evaluator agentic pipeline - LangChain-equivalent orchestration built from scratch - for iterative prompt improvement with cross-model routing across OpenAI, Anthropic, Gemini, and open-source models.
– Implemented quality, latency, and cost-per-output benchmarking via OpenRouter for data-driven model selection, and emitted structured prompt outputs with token budget and version changelog.
– Built a fault-tolerant async backend (BullMQ workers, idempotent retries, Convex persistence) sustaining reliable regression testing without job loss on LLM provider timeouts.

Technical Skills

AI / ML: RAG Pipelines, Generative AI, LLM Applications, NLP, pgvector, BM25, Reciprocal Rank Fusion, Prompt Engineering, Agentic Workflows, Behavioral Memory, LLM Evaluation, TensorFlow, OpenRouter
LLM Providers: Anthropic Claude, OpenAI GPT, Gemini, Groq, Deepgram Nova-2
Languages: Python, TypeScript, JavaScript, SQL, C++, Java
Backend: FastAPI, Node.js, Express.js, BullMQ, Celery, Redis, REST APIs, WebSockets
Frontend: React.js, Next.js, Tailwind CSS, HTML, CSS
Databases: PostgreSQL (+ pgvector), MongoDB, Convex
Cloud & DevOps: AWS, Cloudflare R2, Git, GitHub, async job pipelines
System Design: Hybrid retrieval, multi-agent coordination, event-driven architecture, snapshot-first serving